Xyrus Lorenz M. Dimacali

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OBJECTIVE

Computer Science student with hands-on experience building full-stack applications in Next.js, TypeScript, PostgreSQL, and Python, seeking a Software Development Internship to build scalable production systems and contribute meaningful, well-tested code.

EDUCATION

Manuel S. Enverga University Foundation, Lucena City, Philippines

Bachelor of Science in Computer Science, Major in Software Engineering — GPA: 1.52 (1.00 highest, PH scale) — 2023 to 2027

AWARDS AND HONORS

- College Scholar, First Year to Third Year (2023 to 2026)

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C++, Dart, Swift

Frameworks and Libraries: React, Next.js, Node.js, NestJS, Flutter, pandas, Streamlit

Databases and Backend: PostgreSQL, Prisma ORM, Drizzle ORM, MySQL, Neon Postgres, Supabase, REST APIs

Tools and Platforms: Git, GitHub, Docker, Figma, VS Code, Vercel, Railway, Claude

PROJECTS

RideShareEU: Priority-Scored Greedy Matching Algorithm for University Carpooling — **February 2026 to Present**

Undergraduate thesis — In active development

Stack: Next.js, Node.js, PostgreSQL, Prisma ORM, Python, Jest, Postman, Figma — Code: github.com/KlRAAA/rideshare-eu

- Built the full-stack platform on Next.js and PostgreSQL/Prisma, modeling 8 core relational entities across 8 tracked migrations
- Designed a two-stage matching algorithm, called PSGA, that filters carpool candidates by route overlap, time window, and hard constraints, then ranks survivors on a weighted composite score
- Validated the algorithm against 3 baseline methods (first-in-first-out, route-only, random) on a 500-row human-labeled dataset, outperforming all baselines at 0.826 Precision@1

Kodigo: AI Study Buddy — Bilingual AI-Powered Study Platform — **January 2026 to Present**

Personal project — In active development

Stack: Next.js, TypeScript, Tailwind CSS, Neon Postgres, Drizzle ORM, Google Gemini, Groq, and OpenRouter APIs, Vercel — Live: kodigo-ai-study-buddy.vercel.app/

- Designed a fallback routing layer across three AI providers (Google Gemini, Groq, OpenRouter) with automatic model switching and per-user and global rate limits to stay within free-tier usage
- Implemented spaced-repetition flashcard review using the SM-2 algorithm, along with authentication, content moderation, and an installable Progressive Web App (PWA) experience for iOS users

SkillScope: Exploratory Analysis and Machine Learning on the Tech Job Market — **June 2025 to September 2025**

Personal project — Completed

Stack: Python, pandas, scikit-learn, XGBoost, SHAP, Streamlit, GitHub Actions — Live: skillscope-tech-job-market-analyzer.streamlit.app/

- Cleaned and analyzed 10,040 US tech job postings from a 123,849-row LinkedIn dataset, engineering a 166-skill detection pipeline and normalizing salaries with Interquartile Range (IQR) outlier removal
- Trained an XGBoost classifier to predict seniority level, validated with grouped cross-validation by company to prevent leakage, reaching a 0.607 macro F1 score on unseen companies

WebTracker: Privacy-First Chrome Extension for Active Browsing-Time Tracking — **March 2025 to April 2025**

Personal project — Completed

Stack: JavaScript, Chrome Extension (Manifest V3), Chrome Storage API, Node.js — Code: github.com/KlRAAA/WebTracker

- Designed a reconcile-from-ground-truth session model using chrome.storage.session with atomic, single-threaded queuing to prevent race conditions and survive service-worker shutdowns
- Covered domain parsing, midnight session splits, and reconciliation logic with 114 automated tests, and validated 120-second timing accuracy through real Chrome DevTools Protocol testing

CERTIFICATIONS

- Software Engineering Specialization, IBM (2026)
- Machine Learning with Python, IBM (2026)
- Create the User Interface with SwiftUI, Meta (2025)
- Dynamic Programming and Greedy Algorithms, University of Colorado Boulder (2025)
- Flutter and Dart: Developing iOS, Android, and Mobile Apps, IBM (2025)

CHARACTER REFERENCE

Available on request.